



POSTER #12446

CausalVAE as a Plug-in for World Models: Towards Reliable Counterfactual Dynamics

Keep the predictive backbone. Organize its latent variables for intervention-aware reasoning.

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Paper · arXiv

CORE MESSAGE Predicting the factual future is not enough. A world model should remain reliable after a controlled intervention.

1 Why Counterfactuals Break

Latent world models can retrieve factual futures accurately while their representations remain **entangled** and weakly aligned with causal factors. Under a controlled intervention, the latent change may become diffuse, follow the wrong dependency, or alter unrelated variables.

- **Given:** transitions (o_t, a_t, o_{t+1}) .
- **Intervene:** $\text{do}(s_t^{(i,j)}) = s_t^{(i,j)} + \Delta$.
- **Goal:** factual accuracy plus reliable alternative futures.

Requirement: change the intended factor—and nothing else.

Question: Can explicit causal structure be added without redesigning the predictive backbone?

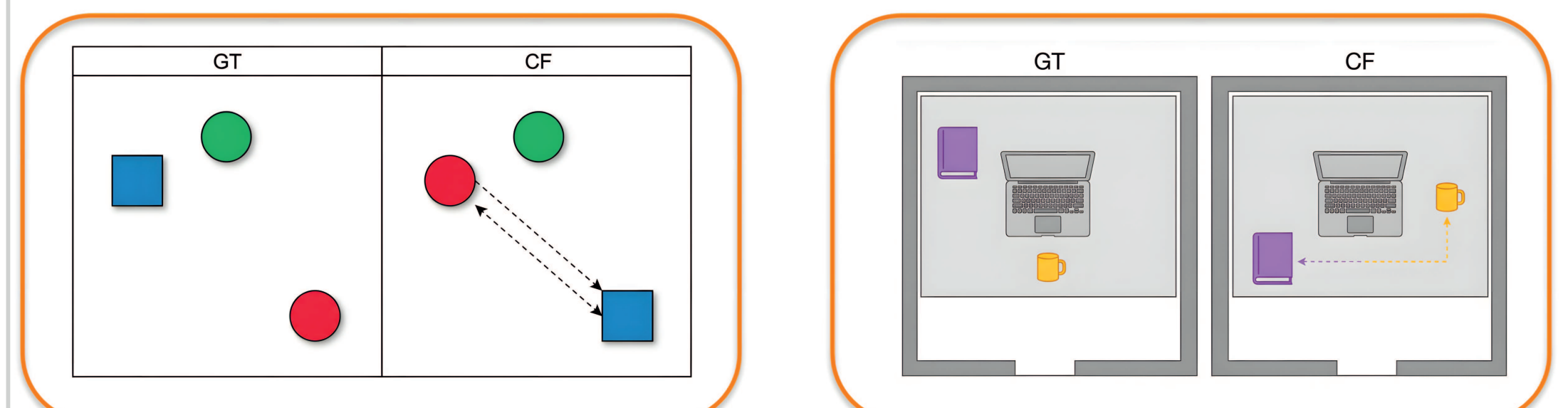
2 Intervention-Centric Evaluation

Each factual tuple is paired with a re-simulated counterfactual target. We intervene at time t , generate o_{t+1}^{cf} , and test whether the model retrieves the correct factual and counterfactual futures.

Counterfactual Task Construction

Intervention: $\text{do}(s_t^{(i,j)}) := s_t^{(i,j)} + \Delta$ ($s_t^{(i,j)}$ denotes the value of the j -th state variable of object i at time t)

Data Construction



Evaluation Pipeline



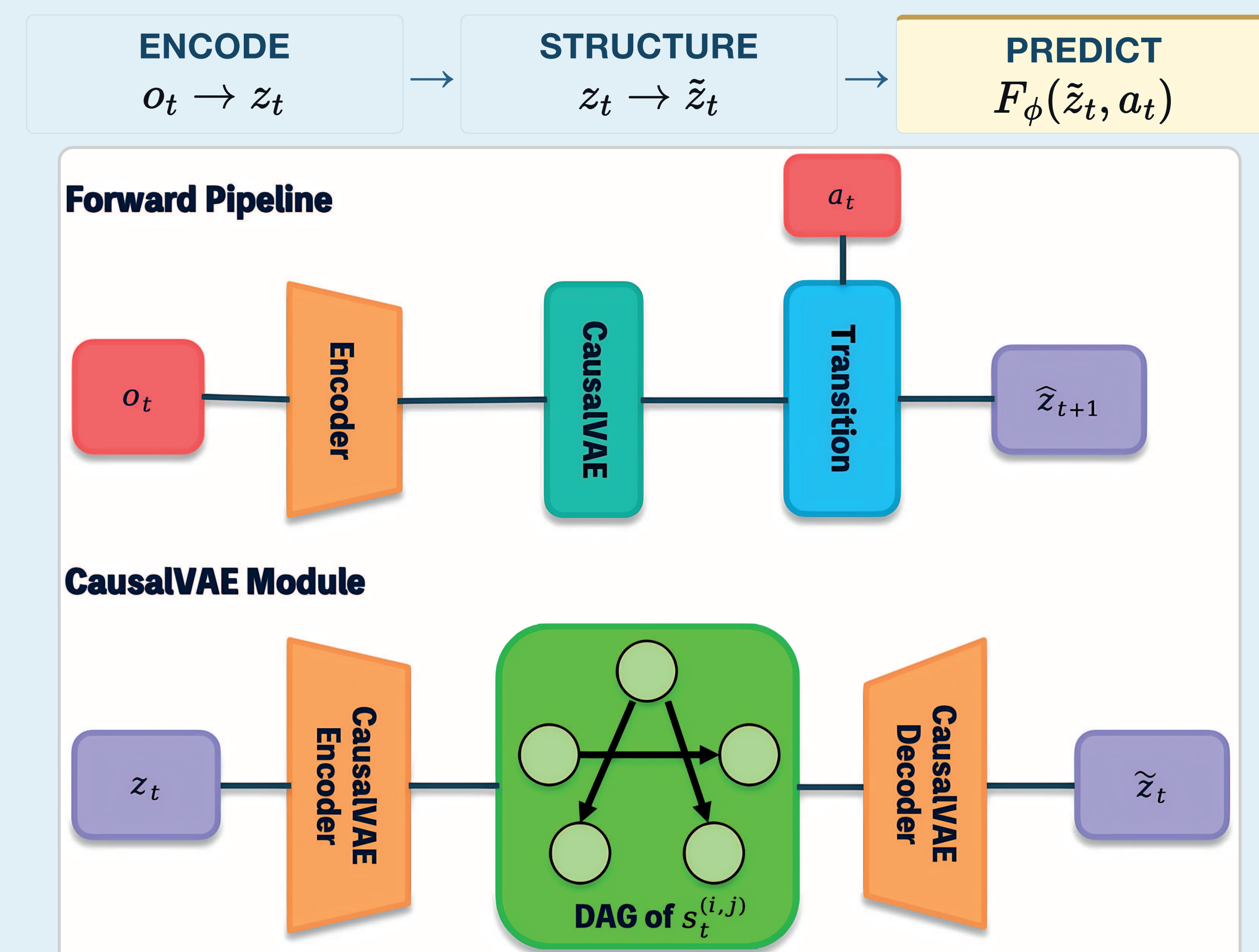
Protocol. Controlled *do*-intervention, re-simulation, then paired retrieval with H@1, MRR, CF-H@1, and CF-MRR.

- Physics** state-level position and velocity interventions
- Shapes / Cubes** object-level pose and push interventions
- Chemistry** mechanism-level interventions

Factual H@1/MRR are evaluated at 1, 5, and 10 steps; CF metrics target the re-simulated alternative future.

3 CausalVAE as a Plug-in ★ KEY

The plug-in operates in latent space. It organizes encoder latents through a learned directed acyclic graph, decodes them back to the host representation, and leaves the base transition architecture unchanged.



Top: inference pipeline. **Bottom:** CausalVAE branch with DAG-structured latent factors.

4 Stage-2 Objective Anatomy

BACKBONE FROZEN

$$\mathcal{L}_{S2} = \lambda_1 \mathcal{L}_{\text{rec}} + \lambda_2 \mathcal{L}_{\text{KL}} + \lambda_3 \mathcal{L}_{\text{align}} + \lambda_4 \mathcal{L}_{\text{DAG}}$$

- rec** retain host latent
- KL** posterior regularity
- align** state anchor
- DAG** acyclic structure

SMOOTH ACYCLICITY

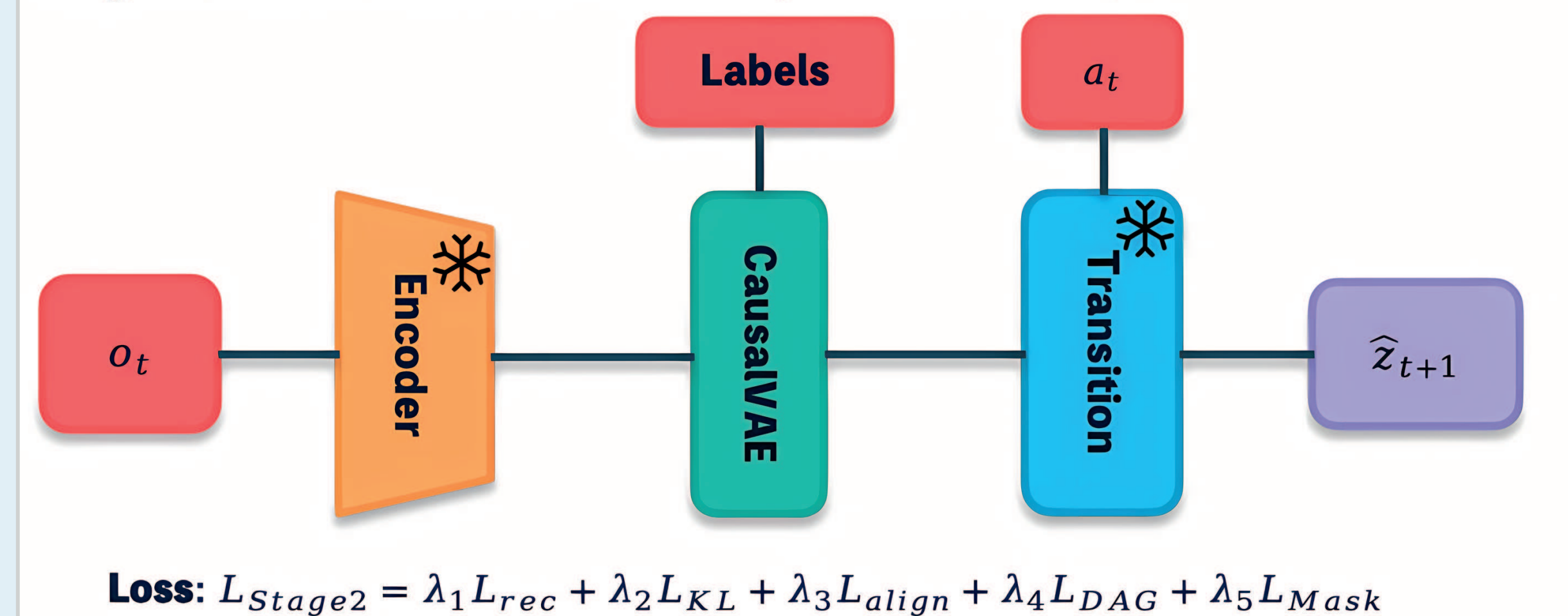
$$\mathcal{L}_{\text{DAG}} = \text{tr}(\exp(A \odot A)) - d$$

Simulator state anchors coordinates only during training; inference requires neither the simulator state nor the alignment head.

5 Three-Stage Training ★ STABLE

- S1 PREDICT** train backbone
- S2 STRUCTURE** freeze backbone
- S3 REFINE** multi-step transition

Stage 2 (Freeze Encoder and Transition, Train CausalVAE)



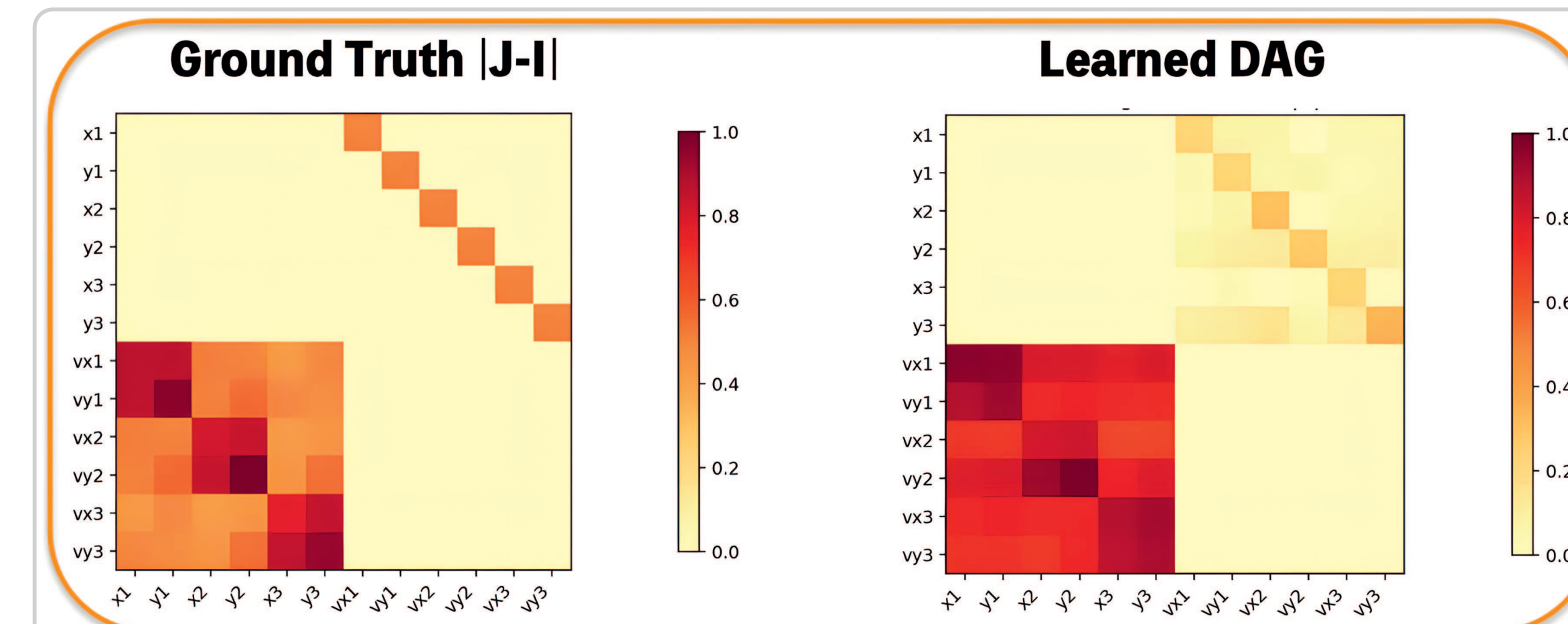
Stage 2: freeze the host encoder and transition, then learn the causal latent module.

ALPHA-GATED FUSION IN S3

$$z_t^{\text{gate}} = (1 - \alpha_t)z_t + \alpha_t \tilde{z}_t$$

6 Interpretable First-Order Structure

On Physics, the learned adjacency follows meaningful interaction trends in the first-order template $A_{\text{GT}} = |J - I|$ derived by local linearization.



$$s_{t+1}^{\text{true}} = F_{\text{true}}(s_t) \approx s_t + \Delta t f(s_t), \quad J := \frac{\partial F_{\text{true}}}{\partial s} \approx I + \Delta t \frac{\partial f}{\partial s}$$

$$A_{\text{GT}} := |J - I|, \quad A_{\text{Learned}} \approx A_{\text{GT}} \text{ (first-order local trend).}$$

Scope: local mechanism-trend correspondence, not recovery of the full nonlinear physical law.

7 Experimental Setting

- **Domains:** Physics 3-body, Chemistry, 2D Shapes, 3D Cubes.
- **Hosts:** AE, VAE, Modular, and GNN dynamics.
- **Metrics:** H@1, MRR, CF-H@1, and CF-MRR.
- **Matched protocol:** identical splits, seeds, optimizer family, and rollout settings within each pair.

8 Physics Counterfactual Results

★ HEADLINE

11→41
GNN-NLL
CF-H@1

+30.5
AE-NLL
gain

+14.0
AE-Contrastive
gain

Physics backbone	Base	+Causal	Gain
AE-NLL	10.5	41.0	+30.5
GNN-NLL	11.0	41.0	+30.0
AE-Contrastive	10.0	24.0	+14.0
Modular-NLL	14.5	22.0	+7.5

The largest gains occur when the host dynamics are intervention-fragile. Across domains, gains remain **benchmark- and backbone-dependent**.

9 Scope and Limitations

Factual retrieval is **usually preserved**, not universally improved. Narrow host latents can discard causal factors before the plug-in sees them. Strongly nonlinear regimes and long-horizon drift remain open problems.

Capacity matters: widening the host latent reverses the Chemistry failure.

Chemistry	Modular-C	+14.44 CF-H@1
2D Shapes	GNN-NLL	+21.75 CF-H@1
3D Cubes	Modular-C	+8.85 CF-H@1

Practical rule: validate the plug-in per backbone and domain; use it when the host latent can carry the relevant factors.

10 Takeaways

IDEA. Causal structure as a plug-in, not a new backbone.

METHOD. Learned DAG plus staged training and gated fusion.

RESULT. Up to +30.5 CF-H@1 points on Physics.

SCOPE. Large but selective gains; factual performance usually retained.